

- 28** In the electron diffraction experiment, the diffraction pattern shows that the de Broglie wavelength of the electron beam is $1.3 \times 10^{-11} \text{ m}$.

What is the potential difference applied to accelerate the electrons from rest?

- A** $5.1 \times 10^{-23} \text{ V}$
- B** $1.4 \times 10^{-15} \text{ V}$
- C** $8.9 \times 10^3 \text{ V}$
- D** $1.4 \times 10^4 \text{ V}$

- 29** X-rays are emitted from a particular X-ray tube with the spectrum D